

ARIMA theory

Navit Dori

November 16, 2025

Is ARIMA with $d = 0$ Equivalent to Linear Regression?

No. ARIMA models with $d = 0$ are **not** equivalent to linear regression.

What ARIMA($d = 0$) Represents

An ARIMA($p, 0, q$) model is an autoregressive–moving-average model:

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \theta_j e_{t-j} + e_t,$$

where y_t depends on past values y_{t-i} and past forecast errors e_{t-j} .

This is a **time-series model**, not a regression on external predictors.

Why It Is Not Linear Regression

A linear regression model has the form

$$y = \beta_0 + \beta_1 x + \varepsilon,$$

in which y is modeled using an external predictor x .

In contrast, ARIMA($p, 0, q$) uses only:

- past values of y (autoregressive part),
- past forecast errors (moving-average part),
- no external regressors.

Therefore it cannot be considered a linear regression model.

When ARIMA Resembles Regression

If $p = 1$, $q = 0$, and $d = 0$, the model becomes

$$y_t = c + \phi_1 y_{t-1} + e_t,$$

which resembles a regression of y_t on y_{t-1} . However, it is still **not** linear regression because:

- parameters are estimated by maximum likelihood, not by OLS,
- residuals must follow an AR(1) structure,
- the model includes stochastic error dynamics typical of time series.

True Linear Regression on Time

A genuine regression model involving time is of the form

$$y_t = \alpha + \beta t + \varepsilon_t.$$

One may detrend the series first and then apply ARIMA, but this is separate from the ARIMA($p, 0, q$) formulation.